

CSIT6910

Independent Project Report

R2T-Based Differentially Private Query Evaluation on TPC-H

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Project Overview

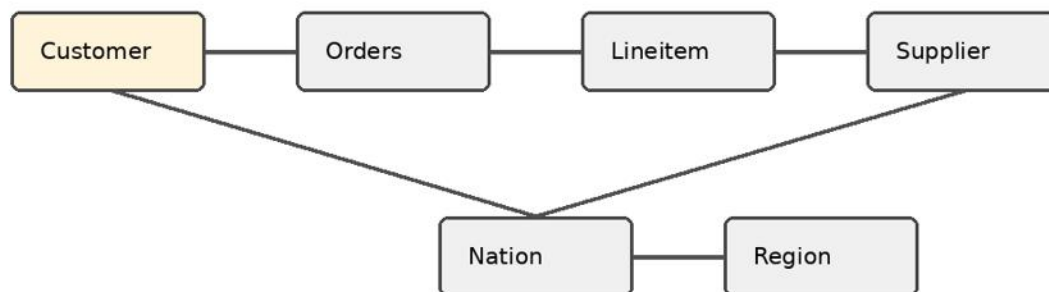
Differential privacy protects the contribution of one private entity when query answers are released. In TPC-H, one customer may reference many orders and lineitems, so a join-and-aggregate query may have large sensitivity. R2T addresses this by evaluating geometrically increasing truncation thresholds and selecting a noisy penalized candidate.

This project implements two concrete TPC-H revenue cases rather than a general SQL engine. Both cases designate Customer as the private relation. The Q3-style case uses Customer, Orders, and Lineitem. The Q5-style case additionally uses Supplier, Nation, and Region.

TPC-H Relations Used by the Two Cases

Q3-style: Customer -> Orders -> Lineitem

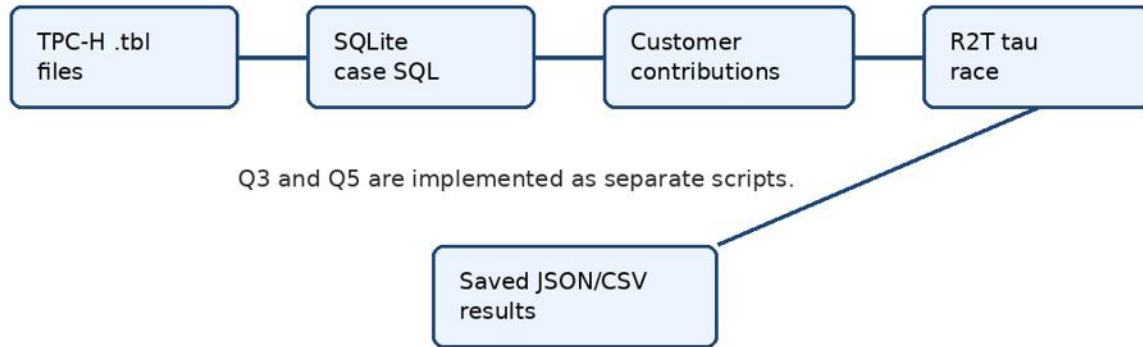
Q5-style: adds Supplier, Nation, and Region



Privacy unit: one Customer and its referencing tuples.

Algorithm Design

R2T Processing Flow



For each case, the program loads the required .tbl files into SQLite and computes each customer's revenue contribution. For a candidate threshold τ , the truncated query value is $\text{sum}(\min(\text{customer_contribution}, \tau))$. Because the implemented cases are self-join-free and protect customers, this is the closed-form equivalent of the corresponding LP truncation.

R2T evaluates $\tau = 2, 4, 8, \dots, \text{GS_Q}$. For each τ it adds Laplace noise with scale $\log_2(\text{GS_Q}) * \tau / \epsilon$, subtracts the R2T penalty term, and returns $\max(0, \text{max noisy candidate})$.

Implemented Queries

Q3-style customer revenue keeps BUILDING customers, orders before 1995-03-15, and lineitems shipped after 1995-03-15.

```
SELECT c.C_CUSTKEY, SUM(l.L_EXTENDEDPRICE * (1 - l.L_DISCOUNT))
FROM Customer c JOIN Orders o ON c.C_CUSTKEY = o.O_CUSTKEY
JOIN Lineitem l ON o.O_ORDERKEY = l.L_ORDERKEY
WHERE c.C_MKTSEGMENT = 'BUILDING'
      AND o.O_ORDERDATE < '1995-03-15'
      AND l.L_SHIPDATE > '1995-03-15'
GROUP BY c.C_CUSTKEY;
```

Q5-style ASIA revenue keeps 1994 ASIA region records and requires customer and supplier nation to match.

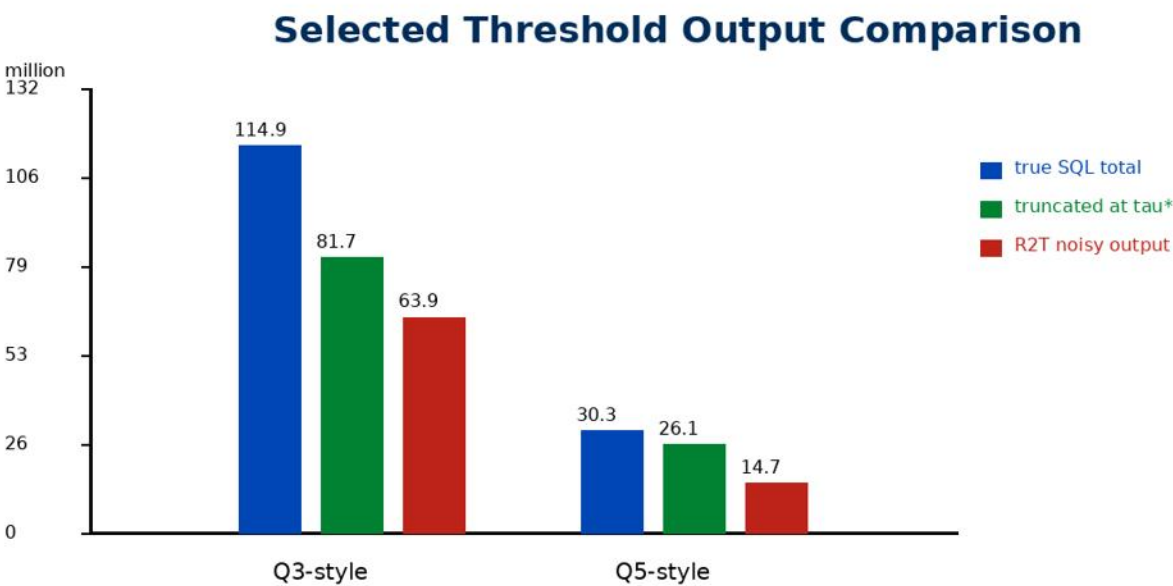
```
SELECT c.C_CUSTKEY, SUM(l.L_EXTENDEDPRICE * (1 - l.L_DISCOUNT))
FROM Customer c JOIN Orders o ON c.C_CUSTKEY = o.O_CUSTKEY
JOIN Lineitem l ON o.O_ORDERKEY = l.L_ORDERKEY
JOIN Supplier s ON l.L_SUPPKEY = s.S_SUPPKEY
JOIN Nation n ON c.C_NATIONKEY = n.N_NATIONKEY
              AND s.S_NATIONKEY = n.N_NATIONKEY
JOIN Region r ON n.N_REGIONKEY = r.R_REGIONKEY
WHERE r.R_NAME = 'ASIA' AND o.O_ORDERDATE >= '1994-01-01'
      AND o.O_ORDERDATE < '1995-01-01'
GROUP BY c.C_CUSTKEY;
```

Experiment Results

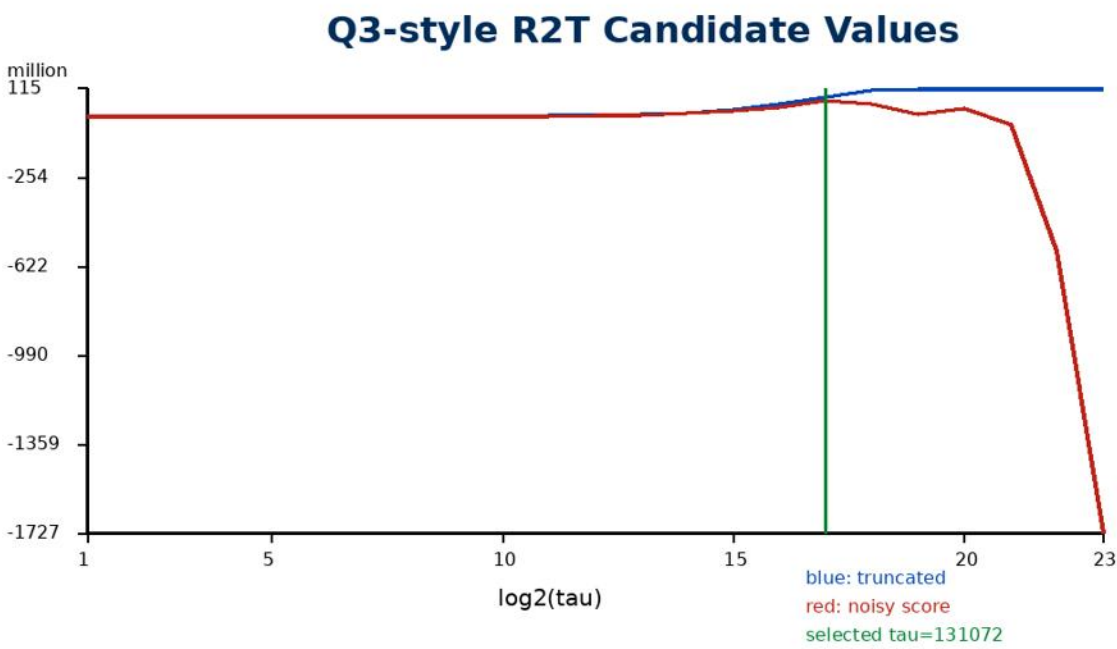
Metric	Q3-style	Q5-style
Contributing customers	904	655
True SQL total	114,904,912.53	30,276,617.68
Max customer contribution	661,967.22	210,018.26
Selected tau	131,072	65,536
Truncated at tau	81,707,181.04	26,148,964.08
Noise scale at tau	3,809,852.89	1,904,926.45
R2T noisy output	63,926,157.02	14,653,924.70

Validation passed for both cases: positive output, monotone truncation, no truncation above the SQL total, and the tau * customer_count bound.

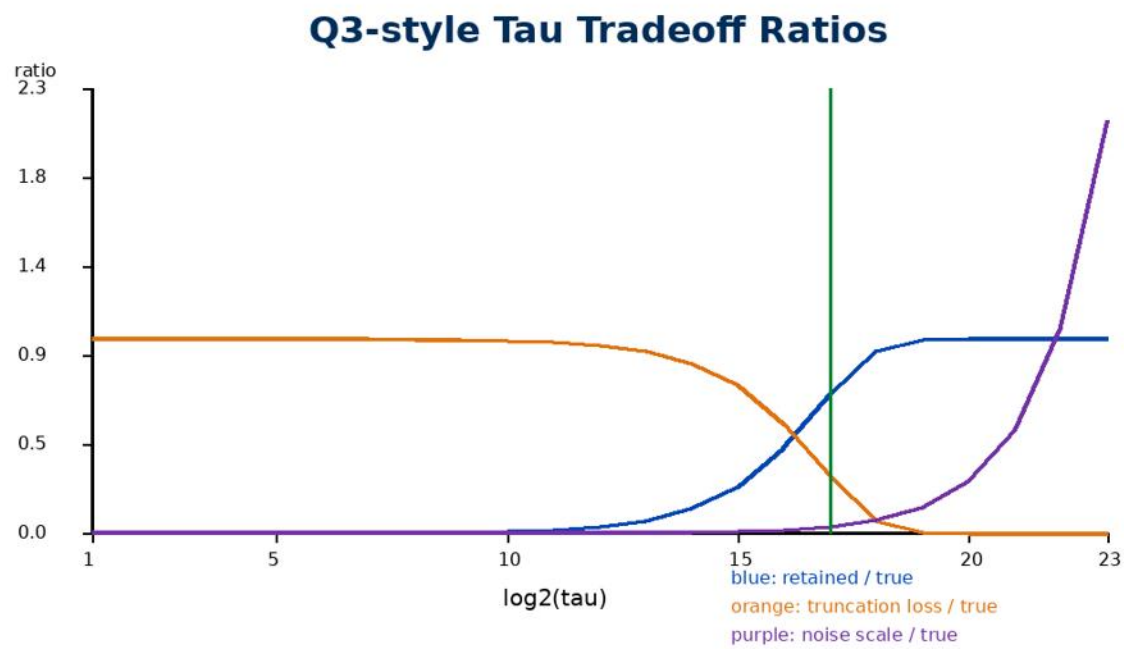
Selected Output Comparison



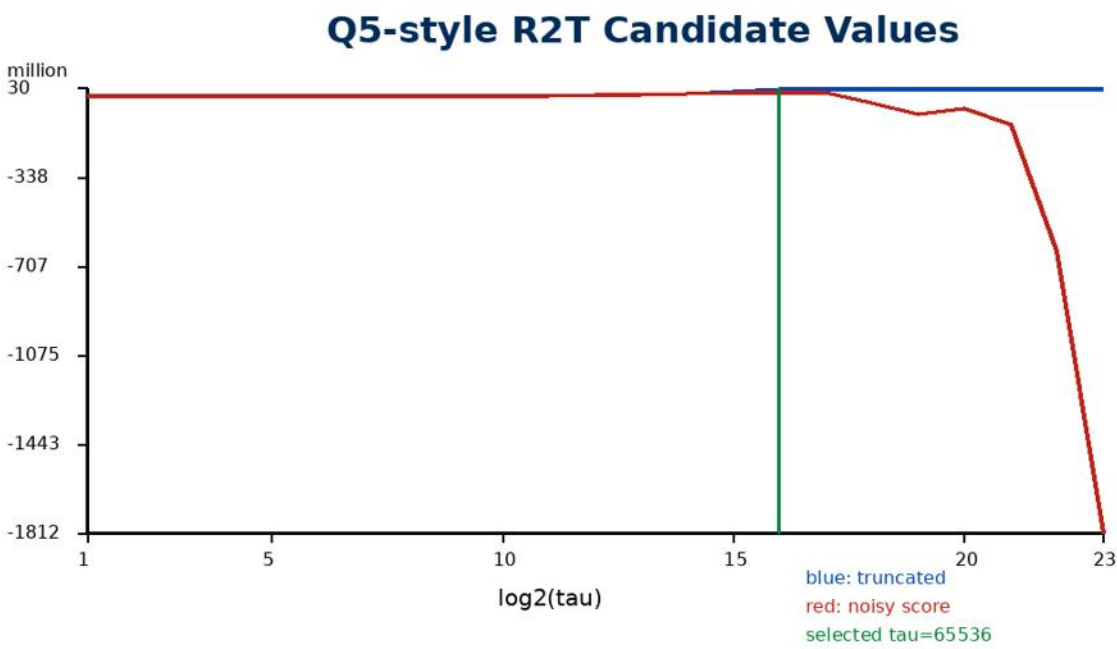
Q3 Candidate Chart



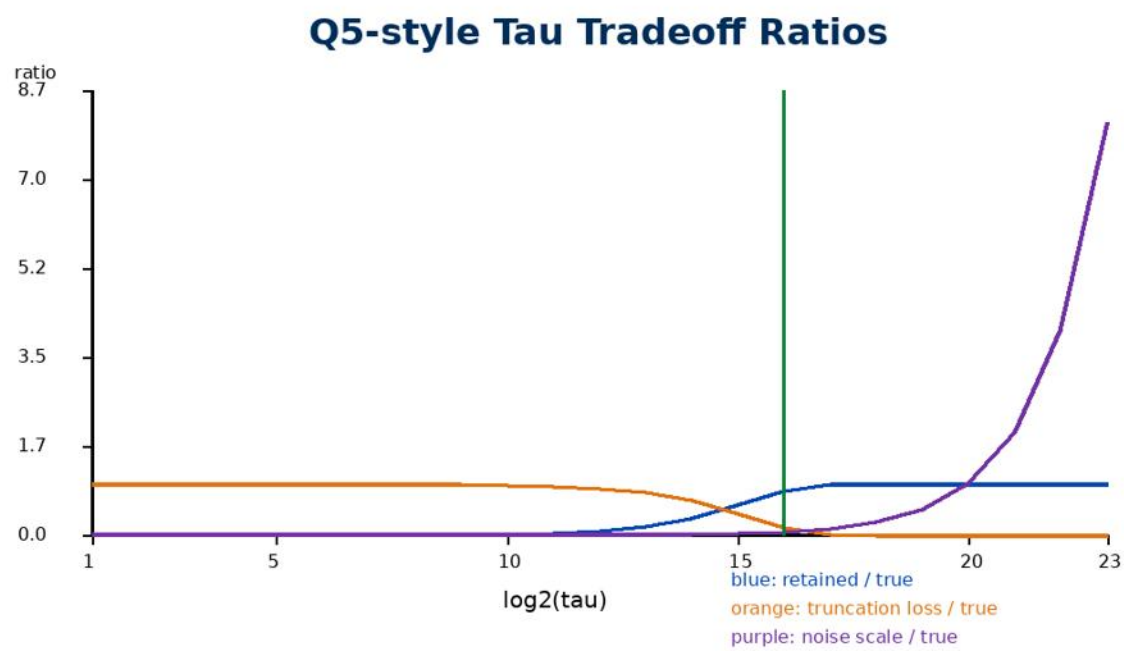
Q3 Tau Tradeoff



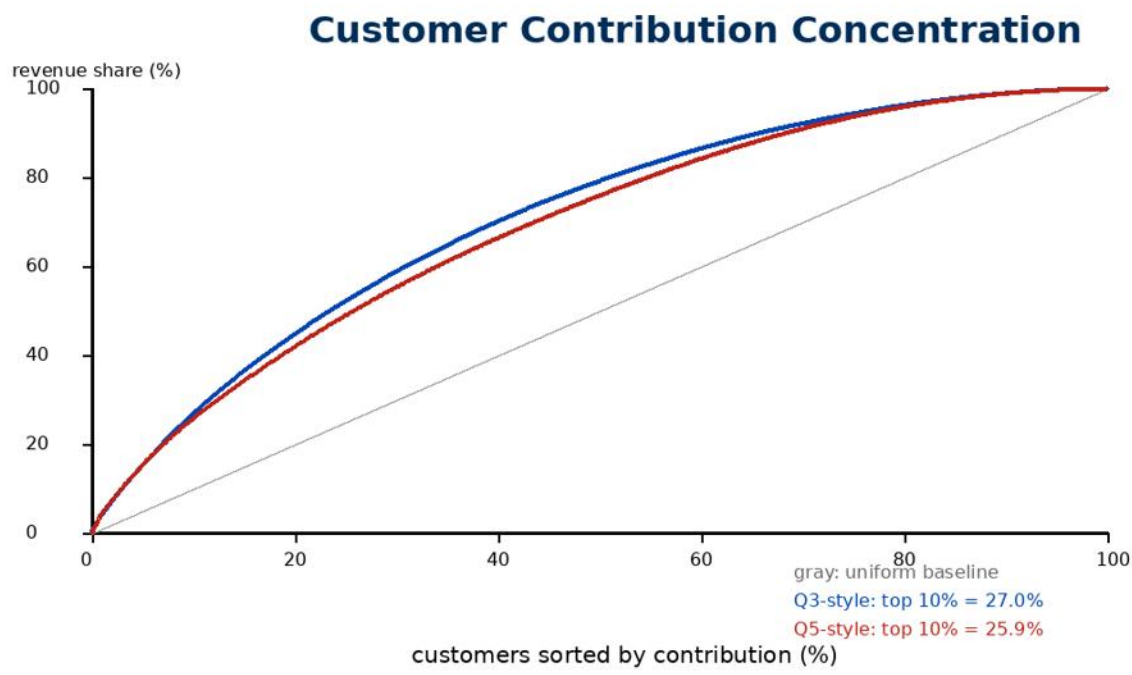
Q5 Candidate Chart



Q5 Tau Tradeoff



Contribution Concentration



Comparison and Conclusion

This project implements two separate Python scripts, saves reproducible outputs, and uses the closed-form truncation for selected self-join-free customer-private queries.

The project demonstrates how R2T can be applied to concrete foreign-key relational queries without implementing a full private SQL engine. Input tables are stored in `data/`, outputs are stored in `results/`, figures are stored in `figures/`, and validation is provided by `code/validate_cases.py`.

Meeting Minutes

1st Project Meeting

Date: Saturday, February 7, 2026

Discussed R2T paper, TPC-H data, and initial project direction.

2nd Project Meeting

Date: Tuesday, March 17, 2026

Reviewed the R2T truncation mechanism and selected two concrete TPC-H cases.

3rd Project Meeting

Date: Tuesday, April 21, 2026

Discussed implementation progress, customer-level contributions, and saved outputs.

4th Project Meeting

Date: Tuesday, May 19, 2026

Reviewed final report, validation results, generated figures, and repository link.